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Research article

A transformer-based generative adversarial learning to detect sarcasm from Bengali text with correct classification of confusing text

Sanzana Karim Lora [a,*], Ishrat Jahan [a], Rahad Hussain [b], Rifat Shahriyar [a], A.B.M. Alim Al Islam [a]

[a] Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh
[b] Department of Bengali Language and Literature, Brac University, Dhaka, Bangladesh

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ABSTRACT

Sarcasm detection research in Bengali is still limited due to a lack of relevant resources. In this context, getting high-quality annotated data is costly and time-consuming. Therefore, in this paper, we present a transformer-based generative adversarial learning for sarcasm detection from Bengali text based on available limited labeled data. Here, we use the Bengali sarcasm dataset 'Ben-Sarc'. Besides, we construct another dataset containing Bengali sarcastic and non-sarcastic comments from YouTube and newspapers to observe the model's performance on the new dataset. On top of that, we utilize another Bengali sarcasm dataset 'BanglaSarc' to further prove our models' robustness. Among all models, the Bangla BERT-based Generative Adversarial Model has achieved the highest accuracy with 77.1% for the 'Ben-Sarc' dataset. Besides, this model has achieved the highest accuracy of 68.2% for the dataset constructed from YouTube and newspaper, and 97.2% for the 'BanglaSarc' dataset.

1. Introduction

Sarcasm is an ironic, biting, harsh, clipping phrase or comment that shows the inverse of what an individual genuinely wishes to say. Sarcasm detection is a category of sentiment analysis topics in which the emphasis is on recognizing sarcasm rather than identifying a general sentiment. It can be difficult because of this deliberate ambiguity, especially in written communication as there are no visible cues like voice inflection or facial expressions. However, the clever and ironic use of sarcasm appears to be employed to convey disdain and subtly provoke offense towards individuals [1,2]. Sarcasm is, therefore, sometimes used by people to cover up or hide their hostile or discriminating intents. They may disguise their harmful or discriminatory sentiments as irony or comedy by using sarcastic remarks to communicate them. This may make it difficult for listeners or readers to understand the underlying meaning behind the speaker's remarks, which may cause misunderstanding or the normalization of detrimental views. For example, "ভাইরে পচাঁ ডিমের রিএক্ট থাকলে একটা দিতাম। *(Oh Bro! If there was a rotten egg reaction, I would have given one.)*" or

* Corresponding author.
E-mail addresses: sanzanalora@yahoo.com (S.K. Lora), ishratjahan039@gmail.com (I. Jahan), rahad.hussain@bracu.ac.bd (R. Hussain), rifat@cse.buet.ac.bd (R. Shahriyar), alim_razi@cse.buet.ac.bd (A.B.M.A.A. Islam).

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"আপনাদের ঠিকানাটা একটু দেন, গালি লিখে একটা চিঠি পোস্ট করবো। *(Please give me your address, I will post a letter with obscenities.)*"- these two examples represent hate speech in guise of sarcasm.

Sarcasm identification studies are available for high-resource languages, including English [3] [4] [5]. However, in the case of low-resource languages, it is a fairly small field of study in Natural Language Processing (NLP). Sarcasm detection on different low resource languages like Indonesian [6], Hindi [7], Czech [8], Japanese [9] are available. Nonetheless, despite being the world's seventh most spoken language, with 240 million native speakers [10], research on sarcasm detection in Bengali is underdeveloped and underutilized. Identifying sarcasm from Bengali text is currently a difficult task for NLP researchers due to limited resources and a paucity of large-scale sarcasm data.

As sarcasm is one of the subclasses of sentiment analysis, there is an area between sarcasm and sentiment where it is very difficult to understand whether it is actually sarcasm or just a simple sentence. It is another challenging and unexplored research area for NLP researchers due to the lack of large-scale annotated data.

Social media has increased in popularity substantially over the previous few decades [11]. According to Kepios[1], there are 4.74 billion social media users in the world in October 2022, accounting for 59.3 percent of the total worldwide population[2]. Placing sarcastic text or graphic material on social networking sites such as WhatsApp, Reddit, Facebook, Twitter, and others has become a new engaging style to avoid blatant pessimism while communicating anything [12].

Deep Learning approaches have grown increasingly popular in NLP in recent years, for example, since they achieve great performance despite depending on relatively basic input representations. Transformer-based architectures, for example, Bidirectional Encoder Representations from Transformers (BERT) [13], give representations of their inputs as a consequence of a pre-training step. These are, in fact, trained on large-scale corpora and then successfully finely tuned over a specific task to get state-of-the-art outcomes in various and diverse NLP tasks. These accomplishments are reached when thousands of annotated examples exist for the final task.

As getting large-scale high-quality annotated data is a very lengthy process as well as costly, a possible solution can be the semi-supervised approach. In a semi-supervised approach, both the advantages of supervised learning and unsupervised learning can be utilized where a small number of labeled data is used.

In this paper, we present such an approach that can detect sarcasm from Bengali text as well as correctly classify confusing text which sometimes creates ambiguity in making decisions about whether it is a sarcastic text or a non-sarcastic one. As preparing large-size high-quality annotated data is very rare for Bengali sarcasm detection, our approach will be such a type that can work best using a few annotated samples.

The rest of this paper is structured as follows. Section 2 outlines the precise research objectives. Section 3 indicates the motivation and practical applications for doing the research. Section 4 discusses prior studies on emotion and sarcasm detection in Bengali and the architectural background. Section 5 demonstrates the detailed architecture of the proposed model. Section 6 then goes into further information about the methodology, which is used to perform the sarcasm detection task. Section 7 determines the correct classification of confusing sarcastic text and sarcasm detection from Bengali text. Section 8 describes the setup and needed parameters to show how our models outperform existing architecture on all datasets. Finally, in Section 9, we bring our research to a conclusion.

2. Research objectives and our contributions

In this section, we are going to clarify our research objectives and pinpoint the specific contributions of this study. It is already mentioned that the detection of sarcasm in Bengali remains a less explored area, despite its vast potential applications. Additionally, the availability of datasets specifically designed for sarcasm detection in Bengali is very limited. Therefore, this research aims to fill this gap by addressing the absence of a suitable model that can effectively identify sarcasm and accurately classify confusing text in the Bengali language. In this study, we propose a transformer-based generative adversarial learning approach, building upon the GAN-BERT architecture [14], which eliminates the need for an extensive amount of high-quality data. Table 1 shows the dummy input-output of the system.

Our primary contributions to this work are highlighted below.

- We introduce a novel method for detecting sarcasm in Bengali text using transformer-based generative adversarial models. Additionally, our approach enables the accurate classification of confusing text with a limited number of labeled examples, all within the same model.
- We curate a new dataset comprising Bengali comments collected from YouTube and newspaper comments. This dataset helps us to expand the range of sarcastic instances available for evaluating the performance of our proposed model.
- We conduct a thorough comparative analysis to identify the best-performing approach. This study involves evaluating the performance of various models and determining the most effective solution for sarcasm detection and classification in Bengali. We evaluate our model rigorously using three datasets which makes our analysis more comprehensive.

[1] https://kepios.com/.

[2] https://datareportal.com/social-media-users.

Table 1
Dummy input-output of our system.

Dummy Input		**Dummy Output**
এতো কষ্ট খালি চোখে দেখা যায় না তাই আমি সানগ্লাস পড়ে দেখা শুরু করলাম *Translation: Such pain cannot be seen with bare eyes so I wore a sunglass.*	**Proposed Model**	Sarcasm
সব খুব সুন্দর কিন্তু নায়ক নায়িকা একটু কথা বললে ভালো হতো, সবই গানের মাধ্যমে হচ্ছে। এটা একটু পালটান তাহলে আরও ভালো লাগবে *Translation: Everything looks beautiful, but if the actors and actresses could speak a little more, it would be even better as everything is happening through songs. Change this a bit, it would be even more delightful.*		Non-Sarcasm
আমি করলেই সমস্যা আর স্যার সাকিব করলে ভালো *Translation: If I do it, then that's a problem, but if Sir Shakib does it, it will be good.*		Non-Sarcasm

3. Motivation behind our study

Social media platforms such as Facebook, YouTube, and Twitter have become major sources in our everyday activities for sharing our views, ideas, and feelings in the present day. Bangladesh has 49.55 million social media users in January 2022[3]. Though Bengali is the seventh most spoken language in the world, sarcasm detection research in Bengali is still unexplored.

People remark on social media to express their thoughts, ideas, and opinions about the content. However, recognizing sarcasm in Facebook Bengali comments is difficult due to the usage of casual language and numerous errors in the content of the comments. Therefore, in this paper, social media Bengali comments are used to identify sarcasm in order to profit from several NLP applications like as market research, opinion mining, advertising, and information classification.

4. Literature review

The main concern of this paper is to detect sarcasm with confusing text with a few labeled data. Therefore, in this section, we discuss two aspects - sarcasm detection and background architecture of working on a few labeled data.

4.1. Emotion and sarcasm detection in Bengali

The rising participation of social media users has an impact on quantitative and qualitative data analysis. Different emotions' corpus on Bengali text was created to enrich the Bengali NLP. 'BEmoC', a corpus containing 7000 Bengali texts gathered from Facebook posts or comments, YouTube comments and different online blog posts were constructed including six basic emotion classes [15]. Emotion detection from Facebook comments using deep learning techniques was studied in [16]. In this work, positive and negative emotions were correctly classified using stacked Long Short-term Memory (LSTM), stacked LSTM with 1D convolution, Convolutional Neural Network (CNN) with pre-trained word embeddings, and, Recurrent Neural Network (RNN) with pre-trained word embedding models where the last one showed the highest accuracy with 98.3%. Bengali social media posts were correspondingly used to detect sentiment in [17]. In this work, LSTM, RNN, and Gated Recurrent Unit (GRU) models were used as well as the hybrid and transformer-based BERT models.

Creating and sharing memes has recently been a popular way to transmit information on social media because memes can spread information in a funny or sarcastic manner. A multimodal hate speech dataset was introduced containing 4158 memes having Bengali and code-mixed captions [18]. A framework was proposed to identify multilingual offense and trolling from social media memes that utilize the weighted ensemble technique to assign weights to the participating visual, textual, and multimodal models [19].

Recently, hate speech detection which is a nearly related area of sarcasm has become a demanding topic among Bengali language researchers. In this regard, the study in [20] presented the construction of a hateful speech detection corpus for the Bengali language on Facebook public pages. The study [21] proposed a model combining BERT and GRU to detect hate speech in Bengali. On the other hand, the study in [22] made use of morphological analysis with a fuzzy classifier and reported a competent performance in detecting Bengali hate speech.

There are just a few contributions in the field of detecting satire, irony, or sarcasm. Satire was discovered in Bengali documents by [23]. They constructed their own Word2Vec model and used the CNN model to obtain an accuracy of 96.4%, although the dataset included insufficient data. [24] identified sarcasm in 41350 Facebook postings by taking into account public replies, interactive comments, and image attachments. To identify sarcasm from photos, they used machine learning techniques and a CNN-based model. Despite the fact that the dataset is large enough, the annotation procedure should have gotten additional attention.

[3] https://datareportal.com/reports/digital-2022-bangladesh.

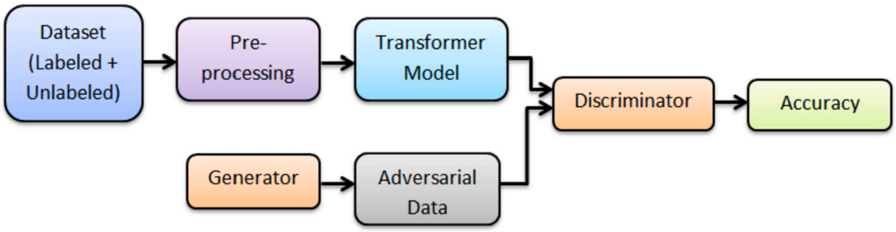


Fig. 1. General structure of the proposed method.

The study in [25], proposed an irony dataset in Bengali comprising 1500 tweets. They used traditional machine learning classifiers such as Random forest, K-Nearest Neighbour (KNN), etc., for irony detection. However, later sarcasm detection using Bengali text was proposed in [26]. A large-scale self-annotated corpus 'Ben-Sarc' containing 25,636 Facebook comments was constructed and proposed a comprehensive technique for employing several models of conventional machine learning, deep learning, and transfer learning to identify sarcasm from text using the 'Ben-Sarc' corpus.

However, very recently, we found a few more studies [27,28] focusing on sarcasm detection in Bengali text. The study in [27], presented a method for identifying and analyzing sarcasm leveraging Global Vectors (GloVe) to represent words and LSTM. Besides, the study presented in [28], experimented with some of the state-of-the-art NLP techniques, machine learning, and deep learning models and proposed a sarcasm detection model for the Bengali language based on an ensemble model consisting of GRU and CNN. Moreover, the study in [29], used a BERT-based system for sarcasm detection and Local Interpretable Model-Agnostic Explanations to introduce explainability to their system.

4.2. Background architecture

Deep Learning approaches have gained popularity in NLP in recent years by achieving great performance using very basic input representations [30] [31]. Some comparative studies, such as the study in [32] demonstrated the superior performance of deep learning than traditional machine learning models on a large dataset containing more than one million tweets.

Recently, Transformer-based architectures like BERT [13] have gained popularity as these give representations of their inputs as a result of a pre-training stage by training on large-scale corpora and then successfully fine-tuned over a specific task to get state-of-the-art outcomes in various and diverse NLP tasks.

As obtaining high-quality annotated data is very time-consuming and costly, one possible solution is to apply a semi-supervised method [33] to overcome this problem when minimal annotated data is available however a collection of unlabeled sources is possible, to increase generalization capacity.

Typically, in a Generative Adversarial Network (GAN) [34], a 'Generator' is trained to create samples that resemble some distribution of data. This adversarial training procedure is dependent on a 'Discriminator', which aims to discriminate samples of the generator from genuine occurrences.

SS-GANs [35] is a GAN extension in which the Discriminator performs better and gives a class to each example while determining whether or not it was created automatically. On the other hand, in the GAN-BERT [14] model, an SS-GAN viewpoint is included in the BERT fine-tuning procedure where a large number of annotated data is not required. In other words, a Generator generates 'fake' samples that resemble the data distribution, while BERT is utilized as a discriminator.

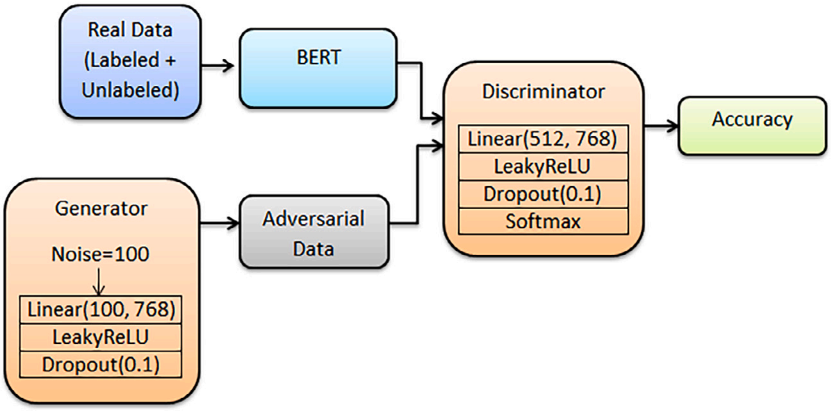
GAN-BERT [14] is used for Bengali text classification [36] by using a few labeled examples. By preserving the architecture of GAN-BERT [14], in their study, they replaced BERT with Bangla BERT and compared the result with Bangla BERT and Bangla ELECTRA with GAN-Bangla-BERT.

As far as we have studied, there is no utilized model for Bengali that can detect sarcasm as well as correctly classify confusing text. In this paper, transformer-based generative adversarial learning is proposed extending the GAN-BERT [14] architecture so that a large number of high-quality data is not required.

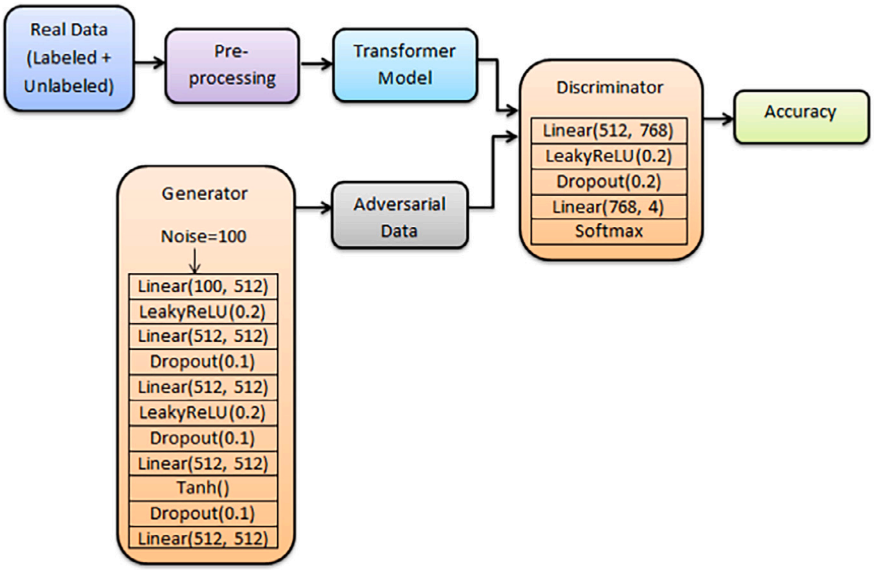
5. Proposed model architecture

In this section, our proposed architecture for sarcasm detection from Bengali text as well as the correct classification of confusing or ambiguous text is described in detail. Fig. 1 represents the general block diagram of our proposed approach and Fig. 2 shows the detailed network architecture of the existing model and our proposed model with all dimensions and parameters with their interconnectivity to demonstrate a comparison. The other steps of the diagram are described in Section 6 in detail. The selection of hyperparameters is described in Section 8.

There are two sub-models - Generator and Discriminator, as we have modified GAN-BERT [14] as our own for the sarcasm detection task. The Generator produces adversarial data and the Discriminator classifies the real data and adversarial data generated by the Generator. The architectures of the Generator and Discriminator are demonstrated in Section 5.1 and 5.2.



(a)



(b)

Fig. 2. Detailed architecture of *(a) existing GAN-BERT* and *(b) our proposed method*. Different sub-blocks are used in Generator and Discriminator blocks to denote the sequence or the structure of Multi-layer Perceptron (MLP).

5.1. Generator

In [14], the Generator was only one hidden layer activated by a LeakyReLU function which takes in input a 100-dimensional noise vector and produces in output a vector as shown in Fig. 2a. However, the network containing only one hidden layer is not sufficient to get an optimum solution as the data is too complex and has high dimensions in the sarcasm detection task. Therefore, we have modified the Generator architecture for our problem.

In modified architecture, the Generator is a multi-layer perceptron. The generator starts with a hidden layer followed by a Leaky Rectified Linear Unit (LeakyReLU) layer as an activation function that takes input as a 100-dimensional noise vector as mentioned in [14] drawn from $N\left(\mu,\sigma^2\right)$. The reason for taking LeakyReLU is to avoid the dying problem caused by classical ReLU [37]. Then another hidden layer with a dropout of 0.1 is added to the architecture to prevent overfitting on the training data. In addition to that, another hidden layer activated by the LeakyReLU function is appended. Then one dropout layer with a dropout rate of 0.1 is added to them. Next, another hidden layer is added and activated by a tanh function. By putting the mean near 0, the tanh function centers the data, which makes learning for the next layer considerably simpler. After that, one dropout layer, and then a hidden layer are added. Finally, the output layer produces a 512-dimensional vector. The output of this Generator is used as the adversarial data in this task. The detailed network interconnectivity is shown in Fig. 2b.

Table 2
Data distribution of 'Ben-Sarc' dataset.

Label	Number of Comments
Sarcasm	12818
Non-Sarcasm	12818
Total	25636

The Generator is anticipated to produce instances that resemble those taken from samples of the actual distribution. According to [35], the Generator should provide data that comes as close as feasible to the statistics of actual data. In other words, the typical example produced by the Generator in a batch should resemble the actual prototype one.

5.2. Discriminator

The discriminator is another Multi-layer Perceptron (MLP) with one hidden layer in [14] activated by a LeakyReLu as shown in Fig. 2a. But it is not enough for our complex task in order to get the best decision boundary. In our case, the discriminator starts with a hidden layer that accepts as input a vector of 512-dimension that may be either adversarial made by the Generator for unlabeled or labeled instances from the real distribution and produces an output of a 768-dimensional vector. This layer is activated by a LeakyReLU. Then, a dropout layer is added with a dropout rate of 0.2. After that, another hidden layer is appended to the network. Finally, the last layer of the Discriminator is a Softmax-activated layer whose output is a $k+1$ vector of logits. The loss function of the Discriminator is defined as: $L_{\mathcal{D}} = L_{\mathcal{D}_{\text{sup.}}} + L_{\mathcal{D}_{\text{unsup.}}}$ where $L_{\mathcal{D}_{\text{sup.}}}$ evaluates the error in selecting the incorrect class from the first k categories to apply to a real example. $L_{\mathcal{D}_{\text{unsup.}}}$ evaluates the error in mistakenly classifying a real (unlabeled) case as adversarial and in failing to identify an adversarial example. The detailed network interconnectivity is shown in Fig. 2b.

5.3. Specialty of our proposed architecture

GAN-BERT [14] is designed for any text classification tasks of NLP. Though sarcasm detection is one type of text classification task, its simple architecture does not give the best result in our task as Bengali sarcastic texts are too complex and confusing in nature. Therefore, we have modified the architecture as described in the previous subsections to improve the accuracy. This is the specialty of our network.

6. Methodology

In this section, we present the overall methodology for sarcasm detection with confusing text using GAN. As shown in Fig. 1, both labeled and unlabeled data are taken in this work and preprocessed. By including a Generator functioning adversely and a Discriminator for categorizing samples, we build the SS-GAN architecture on top of the Transformer language models' design followed by [14]. Then we obtain the accuracy for each transformer model and compare their accuracy. The detailed architecture of the Generator and Discriminator is described in Section 5 and the details of the intermediate steps are described in the next subsections.

6.1. Dataset

'Ben-Sarc' [26], a benchmark dataset for Bengali sarcastic comments on Facebook is used in this work. This is the only dataset for Bengali sarcasm text containing 25636 sarcastic and non-sarcastic comments in a 50-50 ratio. The data distribution of 'Ben-Sarc' is shown in Table 2.

As a semi-supervised approach is followed in our study, we made use of both labeled and unlabeled data. Our experiment is conducted on different percentages of labeled and unlabeled data. Performance according to the percentage of labeled data is shown in Table 8. Among them, data in a ratio of 50-50 for both labeled and unlabeled performs best in the setting mentioned in Table 7.

6.2. Preprocessing

Text preprocessing is a vital task in NLP. There are some basic preprocessing tasks that should be done in NLP research problems. However, in our paper, all preprocessing steps are not mandatory. For example, elongated words frequently include sentiment information [38]. As an example, "খুউউব মজাআআআর *(Veryyyy funnnnny)*" expresses more positive emotion than "খুব মজার *(Very funny)*". Therefore, stemming and lemmatization are not applied to reserve the actual sense of the elongated words as mentioned in [26]. Another text preprocessing step of NLP is stopword removal. However, the stopwords should not always be removed. The removal of stopwords depends largely on the task we are doing and the objective we expect to reach. We choose not to eliminate the stopwords, for instance, if we are building a model to handle the sentiment analysis task. As our task is sarcasm detection and it is a subset of sentiment analysis, stopword removal is not performed in our task. Moreover, Bengali sarcasm comments often contain stopwords. For example, "এও একটা নায়িকা আর কাক ও একটা গায়িকা *(Huh! She is a heroine like a crow is a singer!)*" is a sarcastic text. After removing stopwords, it will be "নায়িকা কাক গায়িকা *(heroine crow singer)*" which does not have any information containing sarcasm. Besides, the punctuation removal step is skipped in our task because the removal of punctuation marks has a negative

Table 3
Preprocessing steps (usual case vs our case).

Preprocessing tasks of NLP	Usual Case	Our Case	Reason for Excluding the Step
Stemming/ lemmatization	yes	no	Elongated words often contain some sentiment information
Removing stopwords	yes	no	Bengali sarcastic comments use stopwords widely
Tokenization	yes	yes	
Removing Punctuation	yes	no	Punctuation removal has a negative impact on overall accuracy
Removing duplicate texts, emoji, links	yes	yes	
Replacing all symbols, hashtag, special characters ('$', '%', '\n') with a single space	yes	yes	
Replacing multiple punctuation(..., !!!) with a single one	rare	yes	

impact on overall accuracy [39]. However, repetitive punctuations are removed because if we preserve more than one punctuation mark, it won't contain any essential information.

Tokenization, removing duplicate texts, emojis, and links, and replacing all symbols, hashtags, and special characters with a single space is utilized according to the usual text cleaning and preprocessing steps of NLP. A summary of preprocessing in this work is shown in Table 3.

6.3. Transformer language models

One of the most common transformer-based models used for pre-training a transformer [40] is BERT [13]. BERT develops deep bidirectional word representations in unlabeled text based on contextual interactions between words. It builds word-piece embeddings based on its vocabulary. BERT pre-training is done with a masked language model (MLM), which randomly masks terms that the model will estimate and compute the loss on, and a future sentence prediction task, in which the model predicts the next phrase from the current sentence.

Four pre-trained BERT-based transformer language models are utilized in our task available in HuggingFace Transformer's library[4] since they are commonly employed in downstream tasks such as text categorization. The transformer language models are as follows:

Bangla BERT(base) [41] It is a pre-trained Bengali language model that was trained using the Bengali Wikipedia Dump dataset[5] and a vast Bengali corpus from the Open Super-large Crawled Aggregated coRpus (OSCAR)[6]. The architecture of the model is bert-base-uncased, which implies it has 12 layers, 768 hidden layers, 49012 heads, and 110M parameters.

Bangla BERT [42] It is a pre-trained ELECTRA discriminator model with the Replaced Token Detection (RTD) objective which allows fine-tuned models to achieve cutting-edge results on many NLP tasks in Bengali.

Indic-Transformers Bengali BERT [43] It is a BERT language model that has been pre-trained on about 3 GB of monolingual training corpus, the majority of which was obtained from OSCAR[6]. It has obtained cutting-edge performance on text categorization tasks in Bengali.

Multilingual BERT (m-BERT) [13] It is a pre-trained model on 102 languages, including Bengali, from the biggest Wikipedia. We utilized the 'bert-base-multilingual-uncased' model. It features 12 layers with 768 hidden layers, 12 attention layers with multiple heads, and 110M parameters.

XLM-RoBERTa(base) [44] It is a multilingual version of RoBERTa. It has been pre-trained on 2.5TB of filtered CommonCrawl data including 100 languages.

6.4. Proposed model

After the dataset is preprocessed, it is fed into the model for the sarcasm detection task. It is mentioned already that in our study, we used transformer-based generative adversarial learning for sarcasm detection in the Bengali language extending the GAN-BERT [14]. We have modified the architecture of the existing Generator and Discriminator of GAN-BERT in several ways to make it appropriate for sarcasm detection in Bengali. The detailed architecture of our proposed model is described in Section 5.

[4] https://huggingface.co/.
[5] https://dumps.wikimedia.org/bnwiki/latest/.
[6] https://oscar-corpus.com/.

Table 4
Examples of confusing text to detect polarity.

Confusing Text
এরকম পোশাক আমিও পড়তে পারি কিন্তু আম্মু লাত্তি দিবে বলে পড়ি না
Translation: I can also wear the same types of clothes, but my mom would kick me so I don't wear them
আমি করলেই সমস্যা আর স্যার সাকিব করলে ভালো
Translation: If I do it, then that's a problem, but if Sir Shakib does it, it will be good.
চ্যাম্প ২২ গজে আপনাকে বিষণ মিস করছি
Translation: Oh Champ, I'm missing you like a poison in 22 yards.
আমার এতো হাসি পায় কেনো রে
Translation: Why am I getting this much laugh?
ওয়আক থু শাকিব
Translation: Yuck Shakib!

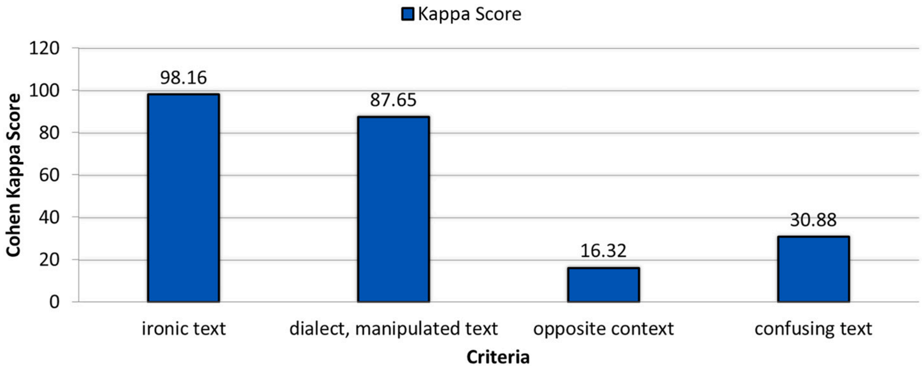


Fig. 3. Inter-annotator agreement of the 'Ben-Sarc' dataset.

7. Correct classification of confusing text and sarcasm detection

In our study, along with the classification of sarcasm detection, we have also explored the classification of confusing texts i.e., texts that seem ambiguous as to whether they express general sentiment or sarcastic intent. In this section, we discuss in detail how our proposed method can detect sarcasm from Bengali texts as well as correctly classify these confusing texts.

7.1. Correct classification of confusing text

Sarcasm detection is a subfield in the broader domain of sentiment analysis. Within the realm of sarcasm detection, some texts create confusion as they exhibit characteristics that blur the line between generic sentiment and sarcastic intent. Therefore, they pose challenges to detect their actual polarity. However, their classification is necessary to enhance the overall accuracy of the model so that it does not get confused when faced with such texts. Table 4 shows some examples of confusing text.

The quality of 'Ben-Sarc' has been measured while constructing it [26]. Fig. 3 shows the percentage of inter-annotator agreement of 'Ben-Sarc' where the Cappa score was computed based on four questions or criteria mentioned in [26]. The criteria are shown in Fig. 3 as ironic text, dialect or manipulated text, opposite context, and confusing text. From Fig. 3, it is clear that only 30.88% data of 'Ben-Sarc' creates confusion to decide, whether it is sarcastic or not. Our goal is to detect sarcasm from Bengali text and classify the confusing one using the same model. Therefore, semi-supervised generative adversarial learning helps us, as it needs a few labeled data. The original architecture of the model is modified in this work to get the state of art performance which was discussed in Section 5.

7.2. Sarcasm detection

Existing GAN-BERT [14] was introduced to work with a few labeled data, for example, almost using 200 samples, and yet it can perform almost the same as a supervised setting which uses a large amount of data. From Fig. 3, it is noticed that 98.16% of the ironic or sarcastic data of 'Ben-Sarc' seems sarcastic according to the inter-annotator agreement of being a text sarcastic where there are several criteria mentioned in [26]. As this portion of data is larger which means more than 12500 and we extend the GAN-BERT, our model can work with this large portion smoothly.

Table 5
A comparison between the performance of existing GAN-BERT and our proposed model on Ben-Sarc dataset for different transformer models with their execution time.

Transformer Model Used in Model	Existing GAN-BERT		Our Proposed Model					
	Accuracy (%)	Execution Time (h:mm:ss)	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)	Execution Time (h:mm:ss)
Indic-Transformers Bengali BERT	65.5	0:16:40	76.1	80.10	65.31	71.95	74.60	0:18:26
Bangla BERT (base)	58.6	0:23:40	71.6	68.82	73.89	71.27	70.33	0:23:49
m-BERT	50.2	0:24:05	70.5	75.24	60.41	67.01	70.34	0:24:20
XLM RoBERTa	50.2	0:28:12	68.9	49.82	100	66.51	50.00	0:26:19
Bangla-BERT	**71.1**	0:23:40	**77.1**	81.80	63.07	71.23	74.57	0:28:29

Table 6
Hyperparameter values which tuned across all models.

Transformer Models	Batch Size	No of Epoch	Dropout Rate	Activation	Optimizer
Bangla BERT(base), XLM-RoBERTa, m-BERT, Indic Transformers- Bengali BERT, Bangla BERT	4, 8, 16, 32, 64, 128	4, 8, 10, 15, 20	0.1, 0.2	ReLU, Tanh, LeakyReLU	Adam, AdamW

Table 7
Hyperparameter setting for best-performed model.

Labeled Data (%)	Epoch	Batch Size	Activation	No of hidden layers (generator)	No of Nodes in hidden layers (generator)	Dropout
50	4	64	LeakyReLU, Tanh	5	100, 512, 512, 512, 512	0.1

8. Experimental evaluation

We conducted rigorous experimentation to validate the performance of our model. In this section, we are going to describe our experimental setup and subsequently present all experiments in detail that are conducted in this work.

8.1. Experimental setup

The Pytorch library is employed to implement all models for training, tuning, and testing. The experiment is carried out on a machine equipped with an Intel Core i5 CPU running at 2.71GHz and 4GB RAM. As we employ Python, we have used Google Colaboratory to create all of the models mentioned in the next sections of this study. All experiments in this section have utilized a Tesla T4 GPU.

8.2. Experiments

Our experiments are divided into two groups. Experiment 1 is focused on the 'Ben-Sarc' dataset experiments. Experiment 2 evaluates the model's performance on another Bengali sarcasm dataset 'BanglaSarc' [45]. Experiment 3 examines the model's performance on a newly created Bengali sarcasm dataset by authors. The accuracy of the models is used to assess their efficacy. The variance of outcomes for each model has been achieved after hyperparameter tuning. For Experiments 2 and 3, all settings are kept the same as in Experiment 1.

8.2.1. Experiment 1: 'Ben-Sarc' dataset

Experiment on existing GAN-BERT architecture and our proposed architecture A comparison between the performance of the existing GAN-BERT model and the proposed model on the 'Ben-Sarc' dataset according to different transformer models is shown in Table 5 with their execution time.

To ensure the quality of our proposed model, we also measure precision, recall, F1-score, and Area Under the Receiver Operating Characteristic (ROC) Curve (AUC). Table 5 also shows the results of the mentioned evaluation metrics. Fig. 4(a-e) and Fig. 5(a-e) depict the confusion matrices and ROC curves, respectively.

Hyperparameter tuning To increase efficiency, hyperparameter tuning is a vital stage for every experimentation. An extensive experiment is conducted for hyperparameter tuning. All values for tuning are shown in Table 6. However, only the hyperparameter setting for a best-performed model with the result is shown in Table 7. The performance of the model according to the percentage of labeled data is shown in Table 8.

Ablation study Before the ablation experiment, we need to choose a baseline model. As existing GAN-BERT has achieved the highest accuracy for Bangla BERT [42], we chose it as the baseline and conducted our ablation experiment. The ablation study for this experiment is shown in Table 9 where * denotes our proposed model.

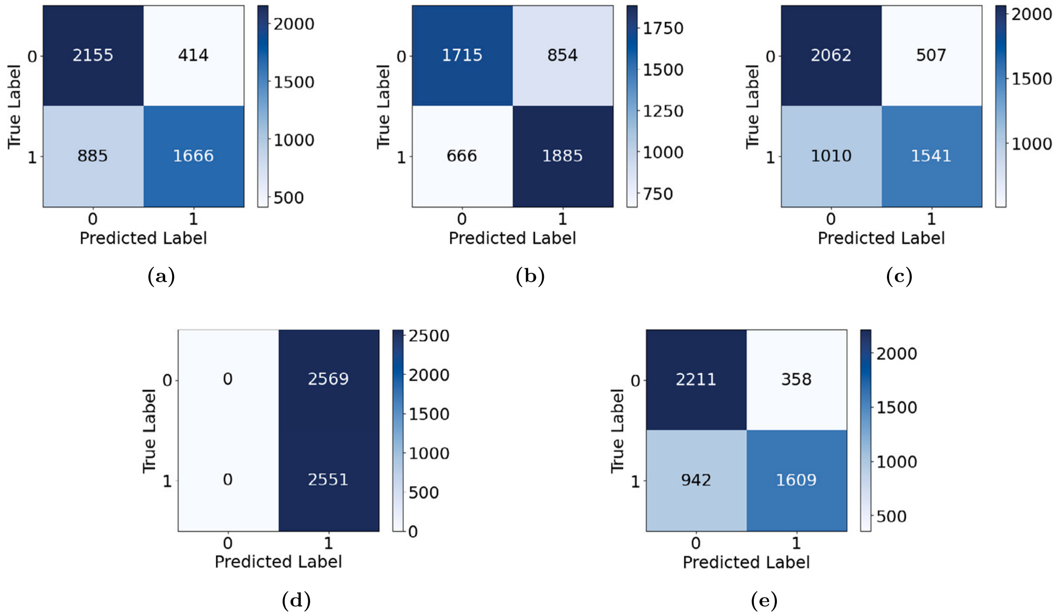


Fig. 4. Confusion matrices of the proposed model on Ben-Sarc dataset using - (a) Indic Transformers Bengali BERT (b) Bangla BERT(base) (c) m-BERT (d) XLM-RoBERTa (e) Bangla BERT.

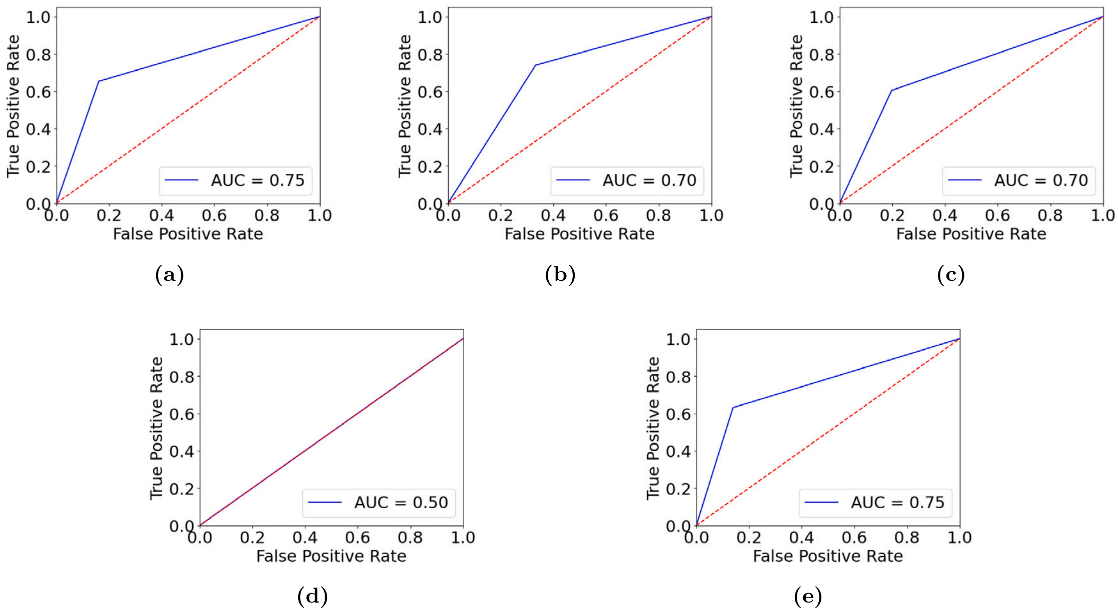


Fig. 5. ROC curve of the proposed model on Ben-Sarc dataset using - (a) Indic Transformers Bengali BERT (b) Bangla BERT(base) (c) m-BERT (d) XLM-RoBERTa (e) Bangla BERT.

Table 8
Accuracy of the proposed model according to the percentage of labeled data and no of epoch.

Percentage of labeled data	**3%**	**30%**	**40%**	**50%**
Epoch	**8**	**8**	**4**	**4**
Model	**Accuracy**	**Accuracy**	**Accuracy**	**Accuracy**
Proposed Model (using Bangla-BERT(base))	62.7	67.3	71.4	71.6
Proposed Model (using Indic-Transformer- Bengali-BERT)	70.4	75.6	75.1	76.1
Proposed Model (using m-BERT)	50.2	62.3	70.1	70.5
Proposed Model (using XLM-RoBERTa)	50.1	51.2	52.2	68.9
Proposed Model (using Bangla-BERT)	72.9	76.4	77.0	77.1

Table 9
Performance of baseline and ablation.

Baseline Model	Accuracy (%)
GAN-Bangla-BERT (1 hidden layer + LeakyReLU in generator)	71.1
Ablation Study (in generator)	
GAN-Bangla-BERT (5 hidden layer + LeakyReLU)	75.4
(6 hidden layer + LeakyReLU)	73.6
(4 hidden layer + LeakyReLU + Tanh)	75.0
(5 hidden layer + LeakyReLU + Tanh)*	77.1
(3 hidden layer + LeakyReLU)	75.8
(2 hidden layer + LeakyReLU)	74.7

Table 10
Data distribution of 'BanglaSarc' dataset.

Label	Status/Comments
Sarcasm	1951
Non-Sarcasm	3159
Total	5112

Table 11
Performance of our proposed model according to Accuracy, Precision, Recall, F1-score, and AUC on BanglaSarc dataset with their execution time.

Model	Transformer Models Used in Our Proposed Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)	Execution Time (h:mm:ss)
Our Proposed Model	Indic-Transformers Bengali BERT	97.1	98.31	91.33	94.69	95.15	0:04:01
	Bangla BERT (base)	94.5	90.95	95.01	92.94	94.40	0:04:58
	m-BERT	94.8	94.45	89.50	91.91	93.02	0:04:48
	XLM RoBERTa	81.0	87.08	88.45	87.76	89.91	0:05:21
	Bangla-BERT	**97.2**	96.35	97.11	96.73	97.60	0:04:28

8.2.2. Experiment 2: 'BanglaSarc' dataset

A single dataset is insufficient to prove the model's reliability. For this reason, 'BanglaSarc' [45], another dataset containing Bengali sarcastic text is utilized in this section. Data preprocessing is kept as same as experimented in Section 6.2.

Data distribution of 'BanglaSarc' This dataset comprises 5112 comments/statuses and contents gathered from different online social platforms such as Facebook, YouTube, and a few blog sites. The data distribution of 'BanglaSarc' is shown in Table 10.

Performance of proposed model on 'BanglaSarc' The accuracy of the 'BanglaSarc' dataset based on the proposed architecture is shown in Table 11 with the execution time. Table 11 also shows the results of the precision, recall, F1-score, and AUC to ensure the models' quality. Fig. 6(a-e) and Fig. 7(a-e) depict the confusion matrices and ROC curves, respectively.

8.2.3. Experiment 3: newly constructed dataset

To check the model's robustness, another dataset for Bengali sarcasm detection is constructed. As Bengali is a low-resource language, the availability of labeled datasets specifically for sarcasm detection in Bengali is very limited. Therefore, a new dataset seems a valuable addition to the literature. Besides, it would allow us to expand the range of sarcastic instances available for training and testing our model. This dataset is constructed from comments collected from YouTube and newspapers whereas 'Ben-Sarc' was constructed using data collected from Facebook only and 'BanglaSarc' from a variety of sources. Therefore, this dataset would capture a broader spectrum of sarcastic expressions, which could improve the model's ability to generalize and perform accurately in real-world scenarios.

Moreover, GAN-BERT [14] claims that it can work like a supervised setting with a few labeled examples even with 200 labeled examples. As we modified GAN-BERT, we constructed a dataset of 1000 samples which is comparatively smaller than the previous two datasets-'Ben-Sarc' and 'BanglaSarc'. A dataset of this small size can help us assess the model's performance in scenarios where limited labeled data is available. On top of that, this would provide insights into the model's adaptability and its potential usefulness in situations where acquiring large labeled datasets is challenging. Besides, in the literature, we also find several studies focusing on sarcasm detection which used datasets that are closer to the size of ours. For instance, the study in [46] experimented with around 600 tweets, [25] worked with 1500 tweets, etc.

Below we present the dataset construction process along with the experiments on the proposed model.

Dataset construction process The dataset construction process is explained below. A sample of the constructed dataset is shown in Table 13.

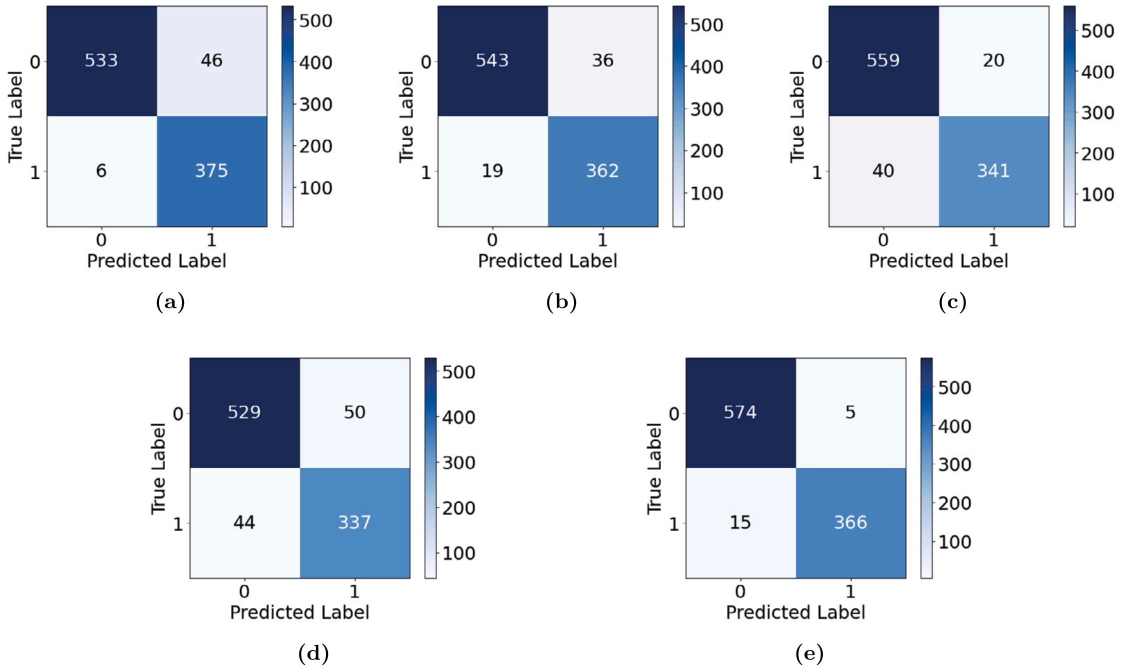


Fig. 6. Confusion matrices of the proposed model on BanglaSarc dataset using - (a) Indic Transformers Bengali BERT (b) Bangla BERT(base) (c) m-BERT (d) XLM-RoBERTa (e) Bangla BERT.

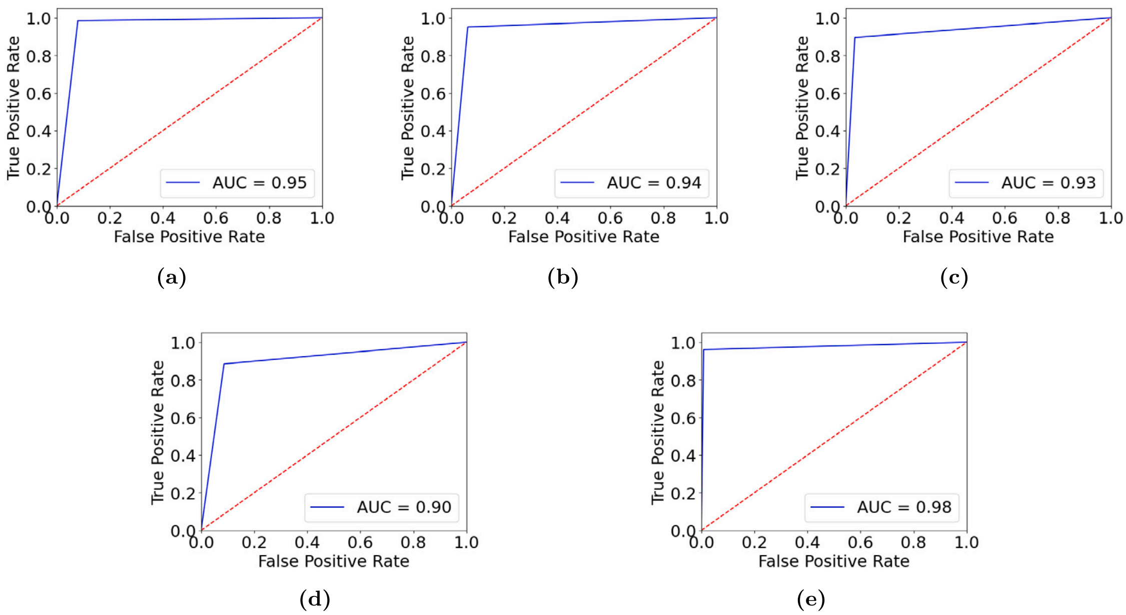


Fig. 7. ROC curve of the proposed model on BanglaSarc dataset using - (a) Indic Transformers Bengali BERT (b) Bangla BERT(base) (c) m-BERT (d) XLM-RoBERTa (e) Bangla BERT.

- **Content Source:** As ‘Ben-Sarc’ contains Facebook comments, we collect YouTube and newspaper Bengali comments both sarcastic and non-sarcastic to create a variety. Five popular Bengali YouTube channels and one newspaper are considered for this purpose. The newly created dataset consists of public comments collected from these sources. We used publicly available data for constructing our dataset and were cautious about not using any private data so that no one’s privacy was hampered. Moreover, the dataset was double-checked and verified by another author.
- **Content Search:** Only the text written in Bengali alphabets is scrapped manually.
- **Annotation Process:** Each text of the dataset is annotated manually using ‘0’ and ‘1’ where ‘0’ means non-sarcastic comments and ‘1’ represents sarcastic comments.

Table 12
Data distribution of newly constructed dataset.

Label	Number of Comments
Sarcasm	500
Non-Sarcasm	500
Total	1000

Table 13
Sample of newly constructed dataset.

Text	Polarity
এদের এত গভীর ভালোবাসা যে লোকজন জানলে তাদের ডিভোর্স করা লাগে *Translation: Their love is so deep that when people know about them, they need to get divorce!*	1
পেশাদার লাইফে আবেগের স্থান নেই। বীরের মা বাবা সঠিক কাজ করেছেন। *Translation: There is/should be no place for emotion in professional life. The parents of Bir has done the right thing.*	0
মাশাআল্লাহ্ চমৎকার প্রতিবেদন ২৬ বছর আগে এই স্কুলে পড়ি বর্তমানে স্কুলের আয়তন বড় হয়েছে ধন্যবাদ *Translation: Wow, excellent report! 26 years ago, I studied in this school, and now it has expanded in size. Thank you!*	0
বিধি তুমি বলে দাও আমি কার, দুটি মানুষ একটি মনের দাবিদার *Translation: Oh God! tell whose am I, Two beings entwined, one heart's outcry!*	1

Table 14
Performance of the proposed model for the newly constructed dataset on different transformer models with execution time.

Model	Transformer Model Used in Our Proposed Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)	Execution Time (h:mm:ss)
Our Proposed Model	Bangla BERT(base)	65.6	85.33	35.71	50.00	62.86	0:01:41
	m-BERT	66.1	69.75	74.11	71.86	64.55	0:01:05
	XLM RoBERTa	60.4	62.86	78.57	69.84	56.79	0:01:21
	Bangla-BERT	**68.2**	86.57	51.78	64.80	70.27	0:01:32

- **Annotation Quality:** The annotation quality of our newly constructed dataset is ensured by the authors of this paper. At first, the first author, who happens to have substantial experience in this domain, annotated the dataset based on her previous experience in this field. Later, the second author verified the annotation using her domain knowledge. Then, we got our dataset validated by an expert. The following paragraph contains the details of the validation process.
- **Dataset Validation by Expert:** After ensuring the quality of the constructed dataset by the authors of this paper, we undertook the task of validating our constructed dataset through collaboration with a language expert. Our expert is a faculty member in Bengali Language and Literature at a reputed university in Dhaka. To eliminate the potential bias, we first removed all our own labels and asked the expert to label independently. Once he was done, we checked and found some labels that needed further clarification. To address this, we organize a collaborative meeting where we, along with our language expert, meticulously examine the contested labeling cases during this discussion, putting forward our own viewpoints. This helped us to clarify underlying contextual complexities and ambiguities, resulting in a consensus-driven choice on final labeling. Cohen's kappa coefficient [47] is used to calculate inter-annotator agreement which determines how well the annotation of the language expert agrees with our annotation by measuring annotator agreement, resulting in a substantial score of 83.4%. This robust kappa score implies a high degree of agreement and alignment between our labeling and the expert's labeling. This is how we got our dataset validated by an expert.
- **Preprocessing:** Preprocessing steps remain as same as was done for 'Ben-Sarc' as discussed in 6.2.
- **Dataset Statistic:** 1000 comments are scrapped where 500 are sarcastic and others are non-sarcastic. The data distribution of the newly constructed dataset is shown in Table 12.

Performance of proposed model on newly constructed dataset The performance of our proposed model on the newly constructed dataset is shown in Table 14 with execution time[7]. Fig. 8(a-d) and Fig. 9(a-d) depict the confusion matrices and ROC curves, respectively.

[7] When we executed the models for our constructed dataset, Indic-Transformers Bengali BERT was publicly unavailable on HuggingFace Transformers library. Therefore, we had to omit this.

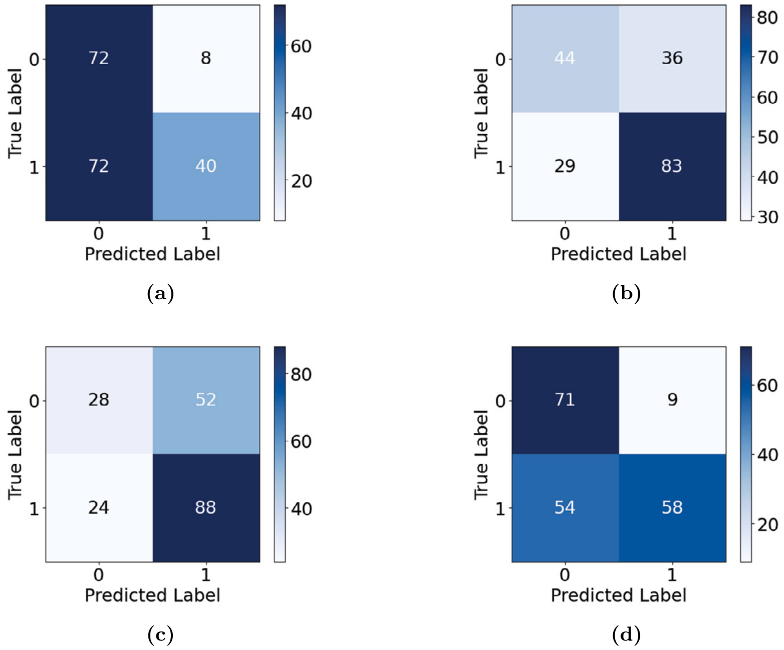


Fig. 8. Confusion matrices of the proposed model on the newly constructed dataset using - (a) Bangla BERT(base) (b) m-BERT (c) XLM-RoBERTa (d) Bangla BERT.

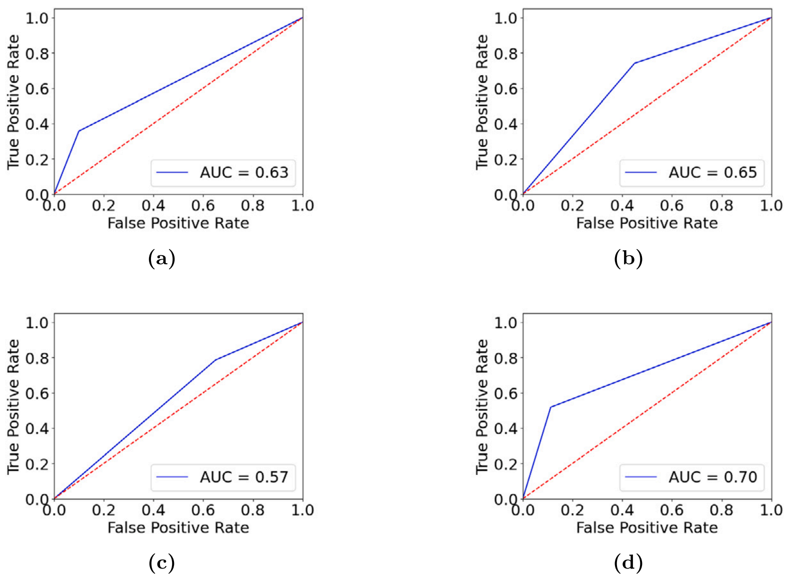


Fig. 9. ROC curve of the proposed model on the newly constructed dataset using - (a) Bangla BERT(base) (b) m-BERT (c) XLM-RoBERTa (d) Bangla BERT.

8.2.4. Statistical analyses

To assess whether the variations among the predictions generated by the five distinct transformer-based models on each dataset are statistically significant or not, we performed several statistical tests. We used the R statistical environment[8] with corresponding packages and default parameters. We used the Friedman test [48] and Quade test [49] to check whether the observed differences in performances across different transformer models were statistically meaningful or simply due to chance. Besides, a pairwise Wilcoxon rank sum test [50] was used to compare models directly.

For all three datasets, the p-value found in both the Friedman test and the Quade test was less than $2.2x10^{-16}$. This implies a significant difference among the predictions of the models. However, these tests do not tell us which groups differ and which do not. Therefore, we performed the Wilcoxon rank sum test to analyze pairwise differences. For all three datasets, the performance of the

[8] http://www.r-project.org/.

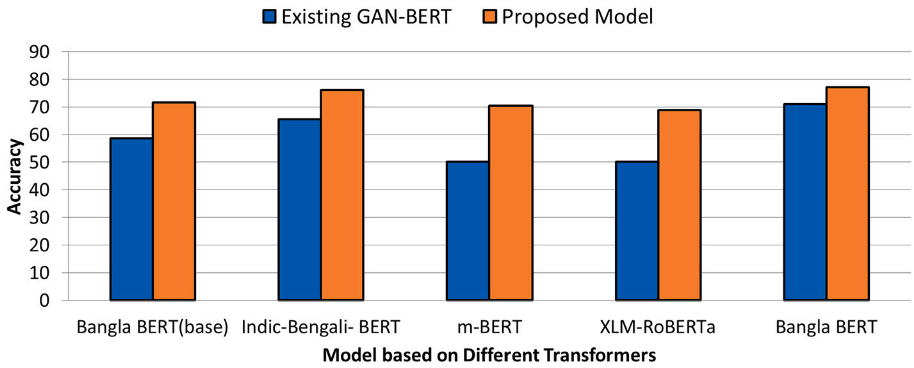


Fig. 10. A comparison between the performance of existing GAN-BERT and the proposed model on 'Ben-Sarc' according to different transformers.

XLM RoBERTa-based model was significantly different than all other models. In the 'Ben-Sarc' dataset, the difference in performance between models based on Bangla-BERT and the Indic Transformers Bengali BERT was not significant ($p = .17$). The same was true for the m-BERT and Bangla BERT (base) based models ($p = .26$). However, all other pairs yielded significant differences ($p < 0.05$). Similarly, the differences in almost all of the pairs were statistically significant for the newly created dataset. On the other hand, for the 'BanglaSarc' dataset, only the difference between the performances of Indic-Transformers Bengali BERT and Bangla BERT (base) was significant. The differences among other pairs were not significant statistically. The reason behind this is all the models except XLM RoBERTa performed very well on this dataset (> 90%) and the performances were much close. Therefore, the pairwise differences did not come out very significant.

8.3. Result analysis and discussion

In this paper, five transformer-based generative adversarial models have been proposed. Among all models, our proposed model based on Bangla BERT [42] achieved the highest accuracy with 77.1% for the 'Ben-Sarc' dataset. It exceeds the result of the existing GAN-BERT. Besides, the second highest accuracy has been obtained by our proposed model based on Indic-Transformers-Bengali BERT with 76.1% which exceeds the result of the existing one. A comparison between the performance of existing GAN-BERT and our proposed model for the 'Ben-Sarc' dataset is demonstrated in Fig. 10.

We conduct the same experiment on 'BanglaSarc' and the newly constructed dataset on the proposed model to observe how the models perform on these datasets. For these two datasets, all settings are kept as same as what is done for 'Ben-Sarc'.

According to the performance shown in Table 11 and Table 14, it can be said that among all models, our proposed model based on Bangla BERT [42] has likewise achieved the highest accuracy with 97.2% and 68.2% for the 'BanglaSarc' and the newly constructed dataset, respectively. The second highest accuracy is from the Indic-Transformers Bengali BERT-based proposed model with 97.1% for the 'BanglaSarc' and the m-BERT-based proposed model with 66.1% for the newly constructed dataset.

As GAN-BERT [14] was proposed to work with a few labeled data, for example, almost using 200 samples it can perform almost the same as a supervised setting using a large data. For this reason, our extended architecture correspondingly performs well for our new dataset using only 1000 samples.

Our proposed model on the 'BanglaSarc' dataset performs almost 20% better than on the 'Ben-Sarc' dataset. First of all, in terms of size, 'Ben-Sarc' is almost five times larger than 'BanglaSarc'. As a result, it is likely that this dataset to capture more varieties of sarcasm and non-sarcasm instances. In contrast, the 'BanglaSarc', being a much smaller one, the diversity is expected to be much less than that of 'Ben-Sarc' and the model finds less difficulty in predicting the correct polarity of the samples. On the other hand, the presence of confusing text seems responsible for the relatively low accuracy as we have kept them intentionally in both 'Ben-Sarc' and the new dataset. We also found almost the same performance on both of these datasets. As mentioned already, confusing texts are those that create confusion about their polarity due to their stance between generic sentiment and sarcastic intent. We intentionally kept them in the datasets as their classification is necessary to enhance the overall accuracy of the model so that it does not get confused when faced with such texts. However, as the name suggests, such texts are also difficult for the model to predict correctly. As there is no such sample present in the 'BanglaSarc', therefore, prediction is somewhat easier on the test samples.

Besides, human evaluation or any experiments that ensure the quality of 'BanglaSarc' is missing where the 'Ben-Sarc' dataset ensures the quality of the dataset in [26]. Therefore, it is not guaranteed that the annotation quality of the 'BanglaSarc' dataset is good enough. This might also play a role behind the performance contrast.

A comparison between the performance of the proposed model on the 'Ben-Sarc' dataset, the 'BanglaSarc' dataset, and the constructed dataset is demonstrated in Fig. 11. From Fig. 11, it is clear that all transformer-based models outperformed on the 'BanglaSarc' dataset in comparison to that of the 'Ben-Sarc' and the newly created dataset. The performance of our proposed model on the 'Ben-Sarc' dataset and the new dataset is nearly the same. Both 'Ben-Sarc' and the newly constructed dataset are constructed following almost the same procedure and both contain a considerable amount of confusing text. Therefore, the model yielding similar performance on them is in line with our expectations. This, in turn, assures the model's ability to generalize well and maintain consistent accuracy levels, regardless of the dataset size. This also suggests that the model is robust and not overly reliant on the

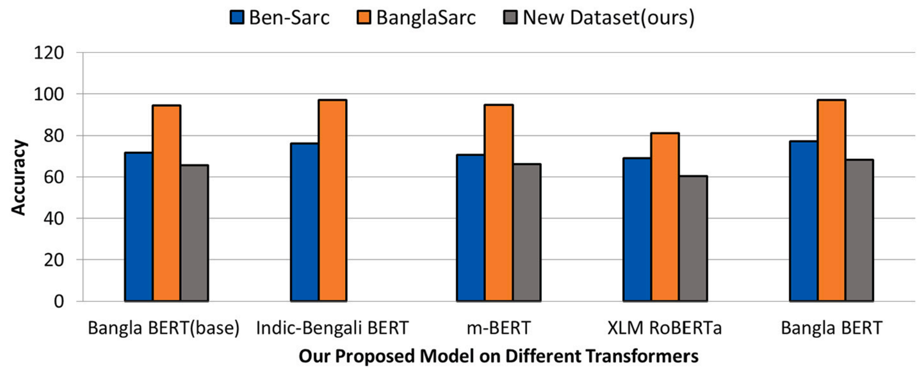


Fig. 11. A comparison between the performance of the proposed model according to different transformers on Ben-Sarc, BanglaSarc, and our newly constructed dataset.

Table 15
Comparison of the performances and other aspects of our study with other related studies.

Author (Year)	Datasets used		Models Used	Accuracy (highest)	Contribution
	#	Size			
Ghosh et al., (2020) [25]	1	1,500	Traditional ML classifiers	67.47%	Irony dataset
Lora et al., (2022) [26]	1	25,636	Traditional ML, DL and Transfer Learning	75.05%	Sarcasm dataset
Apon et al., (2022) [45]	1	5,112	Traditional ML classifiers	89.93%	Sarcasm dataset
Farhan et al., (2023) [28]	1	5,112	GRU + CNN	96%	Sarcasm detection model
Pal et al., (2023) [27]	1	Around 15,000	GloVe + LSTM	91.94%	Sarcasm dataset and detection model followed by sentiment analysis
Anan et al., (2023) [29]	1	5,112	BERT	99.6%	Sarcasm detection model
Our study	**3**	Ben-Sarc - 25,636 BanglaSarc - 5,112 New dataset - 1,000	Modified GAN-BERT	Ben-Sarc - 77.1% BanglaSarc - 97.2% new dataset - 76.0%	Sarcasm dataset and detection model

quantity of training data. It demonstrates the model's capability to effectively learn from limited labeled data, making it practical for scenarios where large labeled datasets may be scarce or costly to obtain. This has practical implications in terms of data collection and annotation efforts, as it allows for more efficient and cost-effective deployment of the model without sacrificing performance.

To facilitate a comprehensive comparison of our work with existing studies related to sarcasm detection in Bengali, below we present a detailed table (Table 15) highlighting the distinguishing aspects of each approach. This table shows that our work stands out from others as we proposed a novel model for sarcasm detection using GAN-BERT along with a new dataset which should be a valuable addition to a low-resource language like Bengali. Besides, in contrast to other studies that tested their models on a single dataset, our work stands out by extensively evaluating our methodology on three distinct datasets which rule out the possibility of the model being overfitted or biased towards a specific dataset. This deliberate inclusion of multiple datasets underscores the robustness and versatility of our approach and demonstrates the effectiveness of our model across a wide range of sarcastic comments in Bengali.

On top of that, another noteworthy aspect of our proposed model in comparison to other studies is that using GAN-BERT as our base model gives us the advantage of training the model with a few samples [14]. This is proved by the consistent performance of the model on the Ben-Sarc dataset (25,636 samples) and the newly created YouTube and newspaper dataset (1000 samples). This characteristic of our model is particularly valuable when obtaining large labeled datasets is challenging.

Moreover, in terms of accuracy, our model outperforms all models of other studies when tested with the dataset 'BanglaSarc' (97.2%) except that of [29] (99.6%). However, as the model presented in [29] is tested with only one dataset, the possibility of overfitting cannot be ruled out. We also included confusing texts in our dataset to make our model robust to handle them which is absent in all other studies. Thus, we anticipate that our study will pave the way for future advancements in the field of Bengali sarcasm detection to a great extent.

9. Conclusion and future work

In this work, we propose a method for sarcasm detection from Bengali text using the transformer-based generative adversarial approach with the correct classification of confusing text with a few labeled examples. Among all models, the Bangla BERT-based proposed model achieves the highest accuracy of 77.1% for the 'Ben-Sarc' dataset. This model even acquires the highest accuracy of 68.2% for the newly constructed YouTube and newspaper comments dataset and 97.2% for the 'BanglaSarc' dataset.

In the future, we will observe the model's performance by increasing the size of the YouTube and newspaper comments dataset and improve the quality of our work by increasing accuracy and considering emojis and emoticons along with the text.

CRediT authorship contribution statement

Sanzana Karim Lora: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ishrat Jahan:** Writing – review & editing, Validation, Project administration, Investigation, Formal analysis. **Rahad Hussain:** Validation, Resources, Data curation. **Rifat Shahriyar:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **A.B.M. Alim Al Islam:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and code will be available on request.

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